Amplification of laser beams counterpropagating through a potassium vapor: The effects of atomic coherence

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We observe amplification of a linearly polarized laser beam propagating through a Doppler-broadened atomic potassium vapor driven by an intense, linear and orthogonally polarized counterpropagating laser beam. The observed gain spectra are well explained by a model that incoporates only the effects of the coherent driving of the atomic dipole moment and not atomic recoil. Our results suggest that the recently reported observations of amplification and lasing using a laser-driven sodium vapor may not arise from the effects of collective atomic recoil. [S1050-2947(97)07210-7]

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I. INTRODUCTION

For many years researchers have exploited the fact that the center-of-mass momentum of an atom changes when it absorbs or emits radiation. This atomic recoil is the basis for laser cooling and trapping of neutral atoms [1] as well as some types of matter-wave interferometers [2] and it gives rise to resonances in the nonlinear spectroscopy of cold atoms [3]. Our understanding of these processes is quite good for weak electromagnetic fields and simple atomic systems but few studies have been performed in the regime where the coupling between the atoms and the radiation field is highly nonlinear or the atomic structure is complex. Recently, Lippi *et al.* [4] and Hemmer *et al.* [5] reported the observation of optical amplification and laser oscillation due to collective atomic recoil in a Doppler-broadened, high-temperature ($\sim$600 K) sodium vapor. They attribute the gain to an atomic density grating (a periodic bunching of the atoms) formed by the aggregate effect of atomic recoil. The modulation depth of the atomic density grating is expected to be small at such high temperatures [6]; however, substantial amplification may occur because of the high atomic number densities. From one point of view, the atoms bunch in response to the spatially-varying dipole force arising from the interference between the pump and probe fields whose modulation depth grows as the probe beam is amplified [7,8]. This mechanism is similar to that occurring in a free-electron laser [9].

In this paper, we investigate the amplification of a weak, linearly polarized laser beam propagating through a Doppler-broadened atomic potassium vapor driven by an intense, linear, and orthogonally polarized counterpropagating laser beam. It appears that atomic recoil effects do not contribute significantly to the observed amplification in our experiments even though the conditions are similar to those described in Refs. [4] and [5]. Rather, our results strongly suggest that the traditional mechanism of the coherent driving of the atomic dipole moment by the laser beams (''coherence'' effects) can explain our observations. This supports our recent theoretical work regarding the possibility that there might be a complex interplay between atomic recoil and coherence effects in the previous experiments [10].

Note that the orthogonal polarizations of the beams used in our experiment preclude the possibility of the two beams interfering, thus eliminating atomic dipole forces as a cause of atom bunching. However, the polarization-gradient force arising from the spatially dependent change in polarization of the optical field is as large in our experiment as the dipole force in the experiments of Refs. [4] and [5]. This is due to the fact that the optical pumping rate of the hyperfine sublevels is comparable to the spontaneous emission rate of the upper state for the intense pump beam used in our experiment [11]. A large polarization gradient force gives rise to a slight spatial redistribution of atoms in different magnetic sublevels and thus can amplify the probe beam through a mechanism similar to that proposed in Ref. [7].

II. OBSERVATION OF LASER BEAM AMPLIFICATION

In the experimental setup the potassium vapor is contained in an evacuated stainless steel cell (length $L=4$ cm) with optical quality windows, into which buffer gases can be introduced. Both the pump and the probe beams are tuned near the $4S_{1/2}\rightarrow 4P_{1/2}$ (D_1) transition in potassium occurring at $\lambda_0=769.9$ nm. The pump laser beam (frequency ω_d, maximum power $P_d=600$ mW incident on the atoms) is generated by an actively-stabilized Ti:sapphire ring laser and collimated to a radius $r_d=370$ μm ($1/e$ field radius) as it passes through the cell, resulting in an intensity of $I_d=2P_d/\pi r_d^2\simeq 280$ W/cm^2. A grating-stabilized diode laser is the source of the probe beam (frequency ω_p, power $P_p=3$ μW) which is collimated to a radius $r_p=200$ μm, resulting in an intensity of $I_p\simeq 5$ mW/cm^2. Both beams are incident on the vapor cell at an angle of $\sim 20°$ with respect to the window normal to prevent multiple reflected beams from entering the interaction region and the angle between the pump and probe beams is $\pi\pm 0.002$ rad. The absolute frequencies of the lasers are calibrated using a Lamb-dip refer-

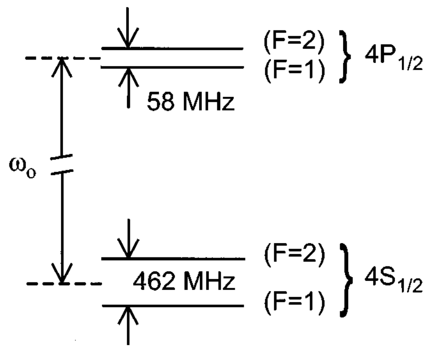

FIG. 1. Energy-level structure of the ^{39}K D_1 transition including the hyperfine structure. The average transition frequency is denoted by ω_0.

ence cell and a low-finesse Fabry-Pérot etalon.

Figure 1 shows the energy-level diagram of the ^{39}K D_1 transition including the hyperfine structure. Throughout our discussion, we reference ω_d to the average frequency

$$\omega_0=\{[\omega_{4P(J=1/2,F=2)}+\omega_{4P(J=1/2,F=1)}]-[\omega_{4S(J=1/2,F=2)}+\omega_{4S(J=1/2,F=1)}]\}/2 \tag{1}$$

of the $4S_{1/2}\rightarrow 4P_{1/2}$ transition, where $\hbar\omega_{4\ell(J=j,F=f)}$ is the energy of the level with quantum numbers (ℓ,j,f). Note that ω_d can be referenced to any specific hyperfine transition using ω_0 and the hyperfine splittings given in Fig. 1.

The reservoir of potassium atoms is contained in a 6-cm-long cold finger attached to the side of the cell, which is heated separately. During the experiment, the cell body is heated to a temperature of 169 °C and the temperature of the cold finger ranges from 129 °C at the tip to 156 °C near the center. This temperature gradient is necessary to prevent potassium from condensing on the cell windows. Since we do not know the precise location of the liquid potassium surface within the cold finger, we are unsure of its precise temperature and hence of the precise value of the atomic number density in the cell. For the range of temperatures between 129 °C and 156 °C, the atomic number density ranges between $\mathcal{N}=3.2\times10^{12}$ and 1.5×10^{13} atoms/cm^3 [12]. This imprecision could be reduced in other, but in our particular case unachievable, experimental conditions.

Our experiment uses potassium rather than sodium (as in Refs. [4,5]) because the hyperfine splitting of the potassium $4S_{1/2}$ state is approximately four times smaller than the splitting of the sodium $3S_{1/2}$ state, thus reducing, but not eliminating, the complexities introduced by the ground-state hyperfine structure. Likewise, we use the D_1 transition, rather than the D_2 transition, to reduce the effects of laser beam self-focusing; and very weak probe-beam intensities are employed to reduce cross-saturation effects such as probe-induced changes in the transverse profile of the pump beam. Working with very low probe beam intensities is made possible by using orthogonally polarized laser beams in combination with polarization filters in front of the detector to reject stray pump beam light scattered from optical surfaces. This choice of conditions increases the likelihood that a simple theoretical model of the interaction can capture the essence of the observed behavior.

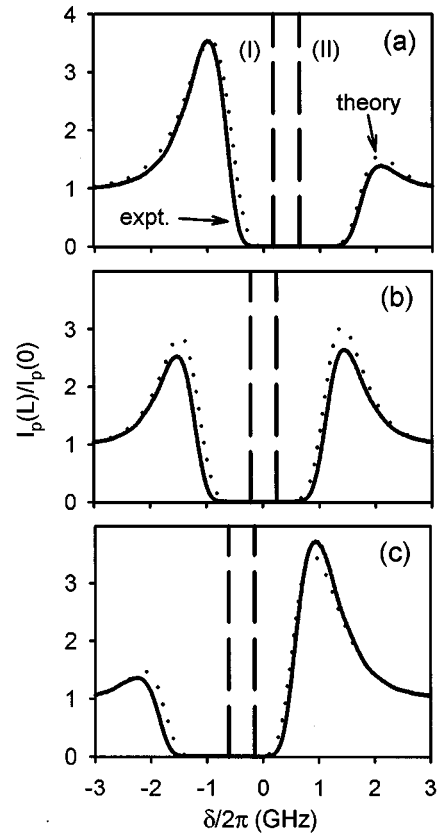

FIG. 2. Gain experienced by the probe beam as a function of pump-probe detuning for three pump detunings: (a) -0.379 GHz, (b) 0.019 GHz, and (c) 0.411 GHz. In each case the solid line is the experimental data and the dotted line is the theoretical prediction with $\mathcal{N}=1.3\times10^{12}$ cm^{-3}. The two vertical dashed lines in each plot indicate the (I) $4S_{1/2}(F=2)\rightarrow 4P_{1/2}(F=1)$ and (II) $4S_{1/2}(F=1)\rightarrow 4P_{1/2}(F=1)$ hyperfine transition frequencies.

To investigate the amplification characteristics of the laser-pumped vapor, we fix the pump frequency, scan the probe frequency, and measure the intensity of the probe beam after the cell. Figure 2(a) shows such an experimental curve as a function of the probe-pump detuning $\delta=\omega_p-\omega_d$ for a pump detuning $\Delta_d=\omega_d-\omega_0=2\pi(-0.376\text{ GHz})$ to the red of the average atomic transition frequency ω_0. For reference, the dashed vertical lines indicate the locations of the $4S_{1/2}(F=1)\rightarrow 4P_{1/2}(F=1)$ and $4S_{1/2}(F=2)\rightarrow 4P_{1/2}(F=1)$ hyperfine transition frequencies. We see that the laser beam is amplified when its frequency is near the low- and high-frequency edges of the Doppler-broadened absorption profile, while it is entirely absorbed on resonance. Our spectrum for $\delta<0$ is rather similar to that obtained by Hemmer *et al.* [see Fig. 3(b) of Ref. [5], note the reversal in the frequency scan] but our observation of amplification for $\delta>0$ is a new result.

III. SIMPLE THEORY OF LASER BEAM AMPLIFICATION

To explore the possible contributions of atomic coherence effects to the observed laser beam amplification, we develop a very simple model of the light-matter interaction that ignores atomic recoil effects. In addition, we make the simpli-

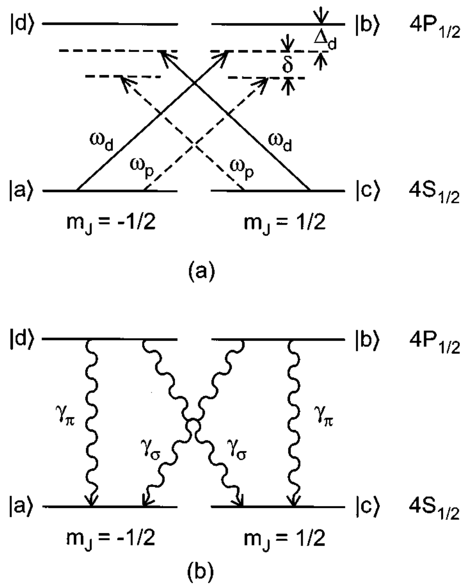

FIG. 3. Energy-level structure for the $J_{1/2}\rightarrow J_{1/2}$ transition. (a) The solid arrows represent the pump laser of frequency ω_d and the dashed arrows represent the probe laser of frequency ω_p, where Δ_d is the detuning of the pump beam frequency from the atomic resonance and δ is the detuning of the probe laser relative to the pump laser. (b) The wavy arrows indicate the decay pathways among the atomic states where $\gamma_\sigma = 2/3T_1$, $\gamma_\pi = 1/3T_1$, and $\gamma_g \approx 2\pi \times 0.11$ MHz arising from the transit time of an atom across the beam.

fying assumption that the spin of the ^{39}K nucleus can be ignored so that the $4S_{1/2}$ ground and $4P_{1/2}$ excited levels each consist of two degenerate Zeeman sublevels with magnetic quantum numbers $m_J = \pm 1/2$ as shown in Fig. 3(a). Thus, optical pumping among the hyperfine levels shown in Fig. 1 is not accounted for in our theoretical treatment. Accounting for these complexities is beyond the scope and intent of our present discussion; our goal is to demonstrate some of the possible characteristics of the interaction in the hope that this will stimulate further experimental and theoretical studies.

Our model uses the density-matrix formalism to treat the interaction between linear and orthogonal polarized plane-wave fields of the form

$$\mathbf{E}(\mathbf{r},t) = \sum_{j=p,d} \boldsymbol{\epsilon}_j A_j(\mathbf{r}) e^{-i(\omega_j t - \mathbf{k}_j \cdot \mathbf{r})} + \text{c.c.}, \qquad (2)$$

and a collection of closed four-level atoms shown in Fig. 3(a) where the ground and excited states each possess total spin $J = 1/2$ with degenerate magnetic sublevels. The plane-wave amplitude of the intense pump (weak probe) field is denoted by A_d (A_p) with wave vector $\mathbf{k}_d = -k\hat{\mathbf{z}}$ ($\mathbf{k}_p \simeq k\hat{\mathbf{z}}$) and polarization unit vector $\boldsymbol{\epsilon}_d$ ($\boldsymbol{\epsilon}_p$), where $\hat{\mathbf{z}}$ is the z-direction unit vector. Taking the quantization axis along $\hat{\mathbf{z}}$, the nonzero dipole matrix elements are given by $\boldsymbol{\mu}_{ba} = \mu_0(\hat{\mathbf{x}} - i\hat{\mathbf{y}})/\sqrt{2}$ and $\boldsymbol{\mu}_{dc} = \boldsymbol{\mu}_{ba}^*$, where $\mu_0 = \mu_{JJ'}/\sqrt{3}$, and $\mu_{JJ'}$ is the reduced dipole matrix element. With this choice of the quantization axis, both the pump and probe beams will interact with the $|a\rangle \rightarrow |b\rangle$ and $|c\rangle \rightarrow |d\rangle$ transitions since they are linearly (but orthogonally) polarized. The laser fields do not interact with the $|a\rangle \rightarrow |d\rangle$ nor the $|c\rangle \rightarrow |b\rangle$ transitions due to the selection rules.

As illustrated in Fig. 3(b), the excited state $|b\rangle$ ($|d\rangle$) decays to the state $|a\rangle$ ($|c\rangle$) at a rate $\gamma_\sigma = 2/3T_1$ and to the state $|c\rangle$ ($|a\rangle$) at a rate $\gamma_\pi = 1/3T_1$, where $T_1 = \hbar c^3/2\mu_0^2\omega_0^3$ is the natural lifetime of the excited states. The atomic dipole moment between $|b\rangle$ and $|a\rangle$ ($|d\rangle$ and $|c\rangle$) dephases in a time $T_2 \equiv 1/\Gamma$. In addition, we incorporate the transit-time broadening of the ground states by assigning a population relaxation rate γ_g between these states. We do not account for the possibility that the coherence between the excited states can be transferred to the ground state coherence via spontaneous emission since this effect is not expected to be important for this system [13]. For simplicity, we treat the interaction to all orders in the pump-beam field strength and to lowest order in the probe field and assume that the pump beam power is not depleted due to transfer of power to the probe beam. However, we properly account for the pump beam absorption due to the atoms.

The field (2) induces a polarization

$$\mathbf{P}(\mathbf{r},t) = \sum_{j=p,d} \mathbf{P}_j(\mathbf{r}) e^{-i(\omega_j t - \mathbf{k}_j \cdot \mathbf{r})} + \text{c.c.} \qquad (3)$$

in the medium that is the source term in the wave equation describing the propagation of the field through the atomic vapor. The polarization amplitudes $\mathbf{P}_j(\mathbf{r})$ are related to the Doppler-averaged, slowly varying off-diagonal density matrix elements of the driven four-level atom at frequency ω_j [14], often called the ''atomic coherences.'' Equation (3) represents a moving polarization grating capable of scattering light. It is intimately connected to a grating in the population difference between the atomic states (the inversions) due to their mutual coupling. An alternate, though entirely equivalent, interpretation of amplification due to coherent effects can be formulated in terms of exchange of photons from the pump beam to the probe beam via scattering from the polarization and inversion gratings formed in the atomic medium. This approach is quite useful under conditions when only the lowest-order nonlinear response of the medium is important, especially for nonresonant fields for which absorption can be neglected [15]. Unfortunately, it is somewhat unwieldy under our experimental conditions due to the highly nonlinear nature of the interaction.

Using Eqs. (2) and (3), we find that the spatial evolution of the intensities of the continuous-wave counterpropagating pump and probe beams is governed by

$$\frac{dI_d}{dz} = \langle \alpha_d(I_d, \Delta_d) \rangle_D I_d, \qquad (4)$$

$$\frac{dI_p}{dz} = -\langle \alpha_p(I_d, \Delta_d, \delta) \rangle_D I_p, \qquad (5)$$

respectively. Note that the probe beam amplification is insensitive to the relative phase between the pump and probe beam (''phase-insensitive amplification'') in contrast to ''wave-mixing'' (also know as ''parametric wave mixing'') processes such as four-wave mixing or parametric up or down conversion. The Doppler-averaged absorption coefficients are given in terms of the coefficients for a collection of atoms moving with velocity v_z along $\hat{\mathbf{z}}$ through the relations

$$\langle \alpha_d(I_d,\Delta_d)\rangle_D=\frac{T_2^*}{\sqrt{\pi}}\int_{-\infty}^{\infty}\alpha_d(I_d,\Delta_d-u)\exp[-(uT_2^*)^2]du, \tag{6}$$

$$\langle \alpha_p(I_d,\Delta_d,\delta)\rangle_D=\frac{T_2^*}{\sqrt{\pi}}\int_{-\infty}^{\infty}\alpha_p(I_d,\Delta_d-u,\delta-2u)\times\exp[-(uT_2^*)^2]du, \tag{7}$$

where $u=kv_z$ is the Doppler shift, $T_2^*=(1/k)\sqrt{m/2k_BT}$ is the Doppler dephasing time, m is the mass of a potassium atom, k_B is Boltzmann's constant, and T is the temperature of the body of the vapor cell. The absorption coefficients for a collection of stationary atoms are given by

$$\alpha_d(I_d,\Delta_d)=(\alpha_0/\mu_0T_2|\Omega_d|)\mathrm{Im}[(\boldsymbol{\epsilon}_d^*\cdot\boldsymbol{\mu}_{ab})\sigma_{ba}^d+(\boldsymbol{\epsilon}_d^*\cdot\boldsymbol{\mu}_{cd})\sigma_{dc}^d], \tag{8}$$

$$\alpha_p(I_d,\Delta_d,\delta)=(\alpha_0/\mu_0T_2|\Omega_p|)\mathrm{Im}[(\boldsymbol{\epsilon}_p^*\cdot\boldsymbol{\mu}_{ab})\sigma_{ba}^p+(\boldsymbol{\epsilon}_p^*\cdot\boldsymbol{\mu}_{cd})\sigma_{dc}^p], \tag{9}$$

where $\alpha_0=(2/3)\mathcal{N}(3\lambda_0^2/2\pi)(T_2/2T_1)$ is the line-center, weak-field absorption coefficient for a linearly polarized beam, $|\Omega_{d,p}|=\sqrt{2}\mu_0|\mathbf{A}_{d,p}(\mathbf{r})|/\hbar=2\mu_0\sqrt{\pi I_{d,p}}/\sqrt{c}\hbar$ is the magnitude of the Rabi frequency, and σ_{ba}^d and σ_{dc}^d (σ_{ba}^p and σ_{dc}^p) are the slowly varying coherences at frequency ω_d (ω_p).

Note that $\langle\alpha_p(I_d,\Delta_d,\delta)\rangle_D$ is a function of I_d, which in turn depends on the spatial location z within the vapor cell in a complex manner due to absorption of the intense pump beam. Hence, we expect that the probe beam intensity will not grow or decay exponentially as it propagates through the cell as would be expected if I_d were constant [10].

The slowly varying coherences are determined through solutions to the density-matrix equations of motion for the four-level atoms illustrated in Fig. 3. For a linearly polarized pump beam with $\boldsymbol{\epsilon}_d=\hat{\mathbf{x}}$, the coherences at the pump beam frequency are given by

$$\sigma_{ba,dc}^d=\frac{w\Omega_{ba,dc}^d}{2(\Delta_d+i/T_2)}, \tag{10}$$

where $\Omega_{ba}^d=\Omega_{ba}^d=|\Omega_d|$ and the population difference between the states $|b\rangle$ and $|a\rangle$ (which is equal to the population difference between $|d\rangle$ and $|c\rangle$) due to the intense pump beam is given by

$$w=\frac{-1/2}{1+|\Omega_d|^2T_1T_2/(\Delta_d^2T_2^2+1)}. \tag{11}$$

The spatial evolution of the pump beam as it propagates through the cell is determined by numerically integrating Eqs. (4) and (6) using Eqs. (8), (10), and (11). This procedure properly accounts for the absorption of the intense pump beam by the atoms. Recall that we use the simplifying assumption that the pump beam power is not significantly depleted due to transfer of power to the probe beam, which is reasonable for the weak probe beam and modest gains in our experiment.

The expressions for the slowly varying coherences at the probe beam frequency are somewhat complicated because the beam is orthogonally polarized ($\boldsymbol{\epsilon}_p=\hat{\mathbf{y}}$) and hence induces population oscillations between the ground states [16]. The probe-induced coherences are determined numerically through solution to the following set of coupled equations

$$[i(\Delta_d+\delta)-\Gamma]\sigma_{ba,dc}^p-iw\Omega_{ba,dc}^p/2-iw_{ba,dc}^p\Omega_{ba,dc}^d/2=0, \tag{12}$$

$$[i\delta-(\gamma_\pi+3\gamma_\sigma)/2]w_{ba}^p-(\gamma_\pi-\gamma_\sigma)w_{dc}^p/2+(\gamma_\sigma-\gamma_g)w_{ca}^p+i[\sigma_{ab}^d\Omega_{ba}^p-\sigma_{ba}^p(\Omega_{ba}^d)^*]=0, \tag{13}$$

$$[i\delta-(\gamma_\pi+3\gamma_\sigma)/2]w_{dc}^p-(\gamma_\pi-\gamma_\sigma)w_{ba}^p/2-(\gamma_\sigma-\gamma_g)w_{ca}^p+i[\sigma_{cd}^d\Omega_{dc}^p-\sigma_{dc}^p(\Omega_{dc}^d)^*]=0, \tag{14}$$

$$[i\delta-(\gamma_\pi-\gamma_\sigma+2\gamma_g)]w_{ca}^p+(\gamma_\pi-\gamma_\sigma)(w_{ba}^p-w_{dc}^p)-i[\sigma_{cd}^d\Omega_{dc}^p-\sigma_{dc}^p(\Omega_{dc}^d)^*]/2+i[\sigma_{ab}^d\Omega_{ba}^p-\sigma_{ba}^p(\Omega_{ba}^d)^*]/2=0, \tag{15}$$

where w_{lm}^p is the population difference between the states $|l\rangle$ and $|m\rangle$ induced by the probe beam, $\sigma_{lm}^d=(\sigma_{ml}^d)^*$, and $\Omega_{ba}^p=\Omega_{dc}^p=i|\Omega_p|$. The spatial evolution of the probe beam as it propagates through the cell is determined by numerically integrating Eqs. (5) and (7) using Eqs. (9) and Eqs. (12)–(15).

We find excellent agreement between the theoretically predicted [dashed line, Fig. 2(a)] and the experimentally measured (solid line) probe beam gain spectrum using the tabulated or experimentally measured parameters ($T_1=25.8$ ns, $T_2^*=0.282$ ns, $T_2=2T_1$, $\gamma_g/2\pi=0.11$ MHz, $\Omega_d/2\pi=1.01$ GHz) and a number density $\mathcal{N}=1.3\times10^{12}$ cm^{-3} (corresponding to a temperature of 115 °C, which is somewhat below the measured range of temperatures). As explained below, we believe that this discrepancy in $\mathcal{N}$ is due to an excitation-dependent reduction in the effectiveness of the atoms, possible due to optical pumping of the potassium ground state.

IV. DEPENDENCE OF LASER BEAM AMPLIFICATION ON EXPERIMENTAL PARAMETERS

Contrary to the experiments of Hemmer *et al.*, we observe amplification for all pump beam detunings in the vicinity of the D_1 transition. For example, Fig. 2(b) (solid line) shows the observed probe beam gain spectrum for $\Delta_d/2\pi=0.019$

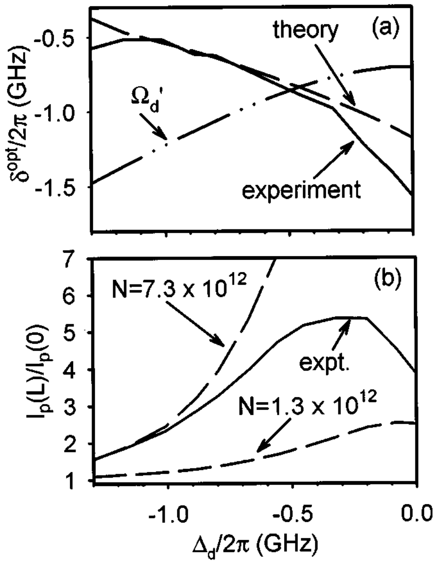

FIG. 4. (a) Probe beam detuning that optimizes the observed (solid line) and theoretically predicted (dash line) amplification as a function of the pump laser detuning. Note that the location of the maximum gain does not follow Ω_d' (the generalized Rabi frequency) indicated by the double-doted-dashed line; rather it scales in the opposite fashion. (b) The observed value of the gain for optimum probe detuning (solid line) agrees well with the theoretically prediced value for large pump detunings with $\mathcal{N}=7.3\times10^{12}$ cm^{-3} (long-dashed line), and reasonably well for small pump detunings with $\mathcal{N}=1.3\times10^{12}$ cm^{-3} (shorted-dash line).

GHz where it is seen that the maximum value of the gain is reduced somewhat in comparison to the previous case and the spectrum is more symmetric. The theoretical prediction (dashed line) tracks these changes where we have used the same parameters as in Fig. 2(a). For $\Delta_d/2\pi=0.411$ GHz [Fig. 2(c)], the spectrum is nearly reversed in comparison to Fig. 2(a). Thus, we find very good agreement between the theoretical predictions and the experimental observations for all three pump detunings by adjusting only the value of the atomic number density (which is held fixed at the adjusted value for all three detunings). These results strongly suggest that the underlying mechanism for the laser beam amplification is the coherent driving of the dipole moment.

The primary differences between our results and those of Hemmer *et al.* and Lippi *et al.* are our observation of gain for both $\delta>0$ and $\Delta_d>0$. While our experiments do not address these differences directly, we speculate that pump and probe beam self-focusing or probe-induced changes in the pump beam transverse profile or propagation direction may play a role since their pump and probe intensities and atomic number densities are quite high. In addition, effects related to optical pumping of the hyperfine levels may play a larger role in their experiments since the hyperfine splitting is significantly larger in sodium than in potassium.

We explore the dependence of the amplification process on the experimental parameters. Figure 4 shows the location and size of the maximum probe gain as a function of Δ_d for $P_d=300$ mW ($\Omega_d/2\pi=0.71$ GHz). We use lower pump beam intensities in this case because the observed gain is somewhat larger. As can be seen in Fig. 4(a), the value of the probe-pump detuning δ^{opt} that optimizes the gain *decreases* in magnitude as the pump beam is tuned further from the atomic transition frequency (solid line) and that our simple theory of the interaction (dashed line) correctly predicts this trend. The optimum probe-pump detuning δ^{opt} is determined by recording a probe beam gain spectrum for a fixed value of Δ_d and measuring the value of δ that give rise to the largest probe beam amplification. If this were a Doppler-free interaction, one would expect that $\delta^{opt}=-\Omega_d'=-\sqrt{\Omega_d^2+\Delta_d^2}$, where Ω_d' is the pump-beam generalized Rabi frequency, as illustrated by the double-dotted-dashed line in Fig. 4(a). Rather, it is seen that the Doppler effect causes the opposite scaling of δ^{opt} on Δ_d. We find that the predicted value of δ^{opt} is not sensitive to the precise value of the atomic number density. Figure 4(b) shows the maximum value of the observed probe beam gain $I_p(L)/I_p(0)$ as a function of Δ_d (solid line) where it is seen that the gain increases initially as the pump beam is tuned away from the atomic resonance and then decreases beyond $\Delta_d/2\pi\sim-0.25$ GHz. The theoretical prediction with $\mathcal{N}=1.3\times10^{12}$ cm^{-3} (long-dashed line, the same number density used in Fig. 2) does not agree well with the observations (recall that the pump beam power is lower in this case). Adjusting the number density to $\mathcal{N}=7.3\times10^{12}$ cm^{-3} (corresponding to $T=140$ °C, well within the range of measured temperatures) improves the agreement between the theory (dashed line) and the observations for large pump detunings. The fact that we obtain better agreement at the larger detunings using a larger value of $\mathcal{N}$ is consistent with an excitation-dependent reduction in the effectiveness of the atoms since fewer atoms are transferred to the excited state for the larger detunings. Mechanisms not included in our theory that might induce such an effect include optical pumping of the hyperfine levels of the potassium ground state, radiation trapping, energy pooling, and velocity changing collisions.

We characterize the dependence of the amplification process on the pump beam power by recording the maximum value of the gain at the optimum values of Δ_d^{opt} and δ^{opt}, as shown in Fig. 5 (solid line). The optimum detunings are defined as the values that give rise to the largest probe beam amplification for a specified pump beam power; they are determined by recording many probe beam gain spectra for a fixed value of P_d and measuring the value of Δ_d and δ that give rise to the largest probe beam amplification. It is seen that the gain rapidly increases for increasing pump beam powers and then decreases above $P_d=360$ mW ($\Omega_d/2\pi=0.78$ GHz). For low pump powers, the theoretical prediction with $\mathcal{N}=7.3\times10^{12}$ cm^{-3} (dashed line) agrees reasonably well with our observations and for higher powers the experimental results are closer to the theoretical predictions with $\mathcal{N}=1.2\times10^{12}$ cm^{-3} (double-dot-dashed line). These observations are also consistent with an excitation-dependent reduction in the effective number density. In Fig 4(b), the optimum pump detuning is plotted as a function of pump power. Our theory captures the correct trend in the data, but is offset from the experimental curve. It is interesting to note that the maximum gain occurs on the opposite side of resonance from the pump laser tuning for low pump powers (i.e., both the theory and experimental curves pass through $\Delta_d^{opt}=0$). A detailed analysis of this transition indi-

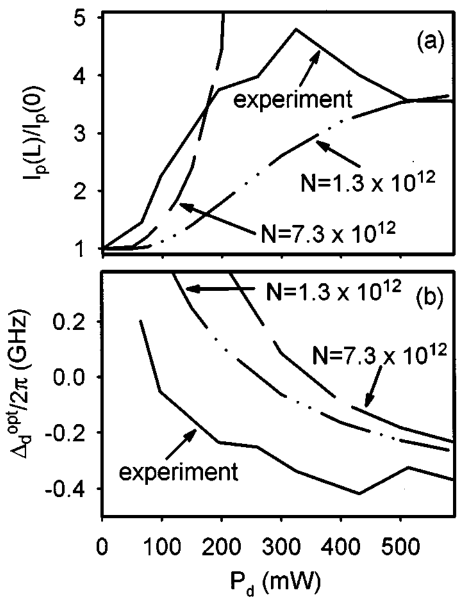

FIG. 5. (a) The observed value of the the gain for optimum pump and probe detuning (solid line) as a function of the pump laser power. The observed behavior agrees well with the theoretically predicted value with $\mathcal{N}=7.3\times10^{12}$ cm^{-3} (double dot-dashed line) at high pump power, while better agreement at low pump power is obtained with $=1.3\times10^{12}$ cm^{-3} (dashed line). (b) The observed value of the pump detuning that optimizes the gain as a function of the pump laser power. The trend of the data is correctly predicted by our simple theoretical model with $\mathcal{N}=7.3\times10^{12}$ cm^{-3} (dashed line) or $\mathcal{N}=1.3\times10^{12}$ cm^{-3} (double-dot-dashed line); however, the curves are consistently offset from our observations.

cates that the dressed-state gain dominates for high pump laser power while Rayleigh gain dominates for lower power (recall that the symmetry of the Rayleigh gain feature with respect to δ is opposite to that of the dressed-state gain and absorption features [10]).

Finally, we introduce a buffer gas into the vapor cell to study the effects of collisions on the interaction. We find that the gain is suppressed by at least a factor of ten for 1.5 Torr of helium at 169 °C. A similar effect is observed at much lower buffer gas pressures (~75 mTorr at 169 °C) for nitrogen. For helium, the dominant mechanisms (in decreasing order of importance) are likely to be a loss of atomic coherence due to dephasing of the coherence (cross section 0.48 nm^2) and nonradiative fine-structure-changing collisions in the potassium excited state ($4P_{1/2}\rightarrow 4P_{3/2}$, cross section 0.26 nm^2) [17–19], while for nitrogen, dephasing of the coherence (cross section 1.77 nm^2), fine-structure-changing ($4P_{1/2}\rightarrow 4P_{3/2}$, cross section 1 nm^2) and quenching ($4P_{1/2}\rightarrow 4S_{1/2}$, cross section 0.35 nm^2) collisions may play a role; hence there is increased sensitivity of the gain to the presence of nitrogen [17,19]. Under our experimental conditions, the collision rates for both gases are comparable to or larger than the spontaneous emission rate. Hence, we expect significant dephasing of the atomic coherence, as well as lifetime broadening of the atomic transition for the case of the nonradiative state-changing collisions. Finally, we note that, although we used the highest quality research-grade gases, we cannot exclude the possibility that some of the reduction in gain that we observed was due to contamination of the cell by residual gases.

V. CONCLUSION

In conclusion, our results suggest that the coherent driving of the atomic dipole moment plays an important role in the amplification of laser beams propagating through a laser pumped atomic medium. New theoretical analyses of these interactions are needed to ascertain the relative importance of the atomic recoil and atomic coherence effects. Important ingredients of an analysis include the strongly saturating nature of the pump beam, the hyperfine splitting of the ground and excited states of the atom, the magnetic substructure of the states, radiation trapping, energy pooling, velocity changing collisions, the closed nature of the atomic system, atomic recoil, and large Doppler widths.

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